Euclid Engine

Free Sample Worksheet

Topic: Attributes & Transformations of Quadratic Functions (Algebra II, Lesson 5.1)

This sample contains two versions of the same worksheet:

Version A — Standard

The worksheet every student receives: 10 problems, standard print.

Version B — Accommodated

The same assignment, regenerated for an accommodation group: fewer problems, larger print, simplified directions, and a word bank. Same math, same standards — different access.

Euclid Engine generates both versions from one click: the teacher picks the topic, and every accommodation group gets its own version with a matching answer key. Answer keys for both versions are included at the end of this packet.

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Transformations of Quadratic Functions

Algebra II · Lesson 5.1 · Version A — Standard

Name: _____ Date: _____ Period: _____

Directions: For each function, state the vertex, state the axis of symmetry, and describe the transformations compared to the parent function $y = x^2$. For the last two problems, write the equation in vertex form.

1. $f(x) = (x - 2)^2 + 5$

2. $f(x) = -(x + 1)^2 - 3$

3. $f(x) = 3(x - 4)^2 + 2$

4. $f(x) = -\frac{1}{2}(x + 5)^2 + 1$

5. $f(x) = (x + 3)^2 - 7$

6. $f(x) = -4(x - 1)^2 + 9$

7. $f(x) = \frac{1}{4}(x - 6)^2 - 2$

8. $f(x) = -(x - 8)^2$

Part II — Write the equation

9. Write the equation: vertex $(-2, 5)$, opens downward, vertically stretched by a factor of 3.

10. Write the equation: vertex $(4, -1)$, opens upward, vertically compressed by a factor of $\frac{1}{2}$.

Transformations of Quadratic Functions

Algebra II · Lesson 5.1 · Version B — Accommodated

Accommodations applied: reduced problem count · larger print · simplified directions · word bank · extra workspace

Name: _____ Date: _____ Period: _____

Word Bank

vertex · axis of symmetry · shift left · shift right · shift up · shift down · stretch (narrower) · compression (wider) · reflection (opens downward)

Directions (simplified):

1. Write the vertex.
2. Write the axis of symmetry.
3. Circle each transformation in the word bank.

1. $f(x) = (x - 2)^2 + 5$

2. $f(x) = -(x + 1)^2 - 3$

3. $f(x) = 3(x - 4)^2 + 2$

4. $f(x) = (x + 3)^2 - 7$

5. $f(x) = -\frac{1}{2}(x + 5)^2 + 1$

6. Write the equation: vertex $(4, -1)$, opens upward, compressed by $\frac{1}{2}$.

Answer Key — Version A (Standard)

For teacher use

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1. $f(x) = (x - 2)^2 + 5$

Answer: Vertex (2, 5); axis x = 2; shift right 2, shift up 5.

2. $f(x) = -(x + 1)^2 - 3$

Answer: Vertex (−1, −3); axis x = −1; reflected over the x-axis (opens downward), shift left 1, shift down 3.

3. $f(x) = 3(x - 4)^2 + 2$

Answer: Vertex (4, 2); axis x = 4; vertical stretch by 3 (narrower), shift right 4, shift up 2.

4. $f(x) = -\frac{1}{2}(x + 5)^2 + 1$

Answer: Vertex (−5, 1); axis x = −5; reflected over the x-axis, vertical compression by ½ (wider), shift left 5, shift up 1.

5. $f(x) = (x + 3)^2 - 7$

Answer: Vertex (−3, −7); axis x = −3; shift left 3, shift down 7.

6. $f(x) = -4(x - 1)^2 + 9$

Answer: Vertex (1, 9); axis x = 1; reflected over the x-axis, vertical stretch by 4 (narrower), shift right 1, shift up 9.

7. $f(x) = \frac{1}{4}(x - 6)^2 - 2$

Answer: Vertex (6, −2); axis x = 6; vertical compression by ¼ (wider), shift right 6, shift down 2.

8. $f(x) = -(x - 8)^2$

Answer: Vertex (8, 0); axis x = 8; reflected over the x-axis (opens downward), shift right 8.

9. Vertex (−2, 5), opens downward, stretched by 3:

Answer: $f(x) = -3(x + 2)^2 + 5$

10. Vertex (4, −1), opens upward, compressed by ½:

Answer: $f(x) = \frac{1}{2}(x - 4)^2 - 1$

Answer Key — Version B (Accommodated)

For teacher use · numbering matches Version B

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1. $f(x) = (x - 2)^2 + 5$

Answer: Vertex (2, 5); axis x = 2; shift right, shift up.

2. $f(x) = -(x + 1)^2 - 3$

Answer: Vertex (−1, −3); axis x = −1; reflection, shift left, shift down.

3. $f(x) = 3(x - 4)^2 + 2$

Answer: Vertex (4, 2); axis x = 4; stretch, shift right, shift up.

4. $f(x) = (x + 3)^2 - 7$

Answer: Vertex (−3, −7); axis x = −3; shift left, shift down.

5. $f(x) = -\frac{1}{2}(x + 5)^2 + 1$

Answer: Vertex (−5, 1); axis x = −5; reflection, compression, shift left, shift up.

6. Vertex (4, −1), opens upward, compressed by ½:

Answer: $f(x) = \frac{1}{2}(x - 4)^2 - 1$